



GlucoTwin

A Type 2 diabetes digital twin that warns of a glucose spike a median 90 minutes before it happens

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Happiest Health Digital Twin Challenge 2026 · Phase 1 submission

Diabetes care in India is still reactive

Quarterly HbA1c tests average away the daily spikes that drive complications

101 M

Indians living with diabetes
(ICMR-INDIAB, 2023)

136 M

more with prediabetes (same
study)

~3 months

typical gap between HbA1c
checks, blind to daily excursions

- Post-meal glucose spikes are common with carbohydrate-heavy Indian meals and late dinners.
- Short sleep and stress quietly raise insulin resistance the next day, and nobody connects the dots.
- CGMs show where glucose is now. Doctors need to know where it is going, and what would change it.

Our focus: predict the next post-meal spike

One condition, one outcome, done well, as the challenge brief asks

Prediction target

- Condition: Type 2 diabetes
- Event: CGM glucose > 180 mg/dL (ADA post-meal target)
- Horizon: any time in the next 2 hours
- Only predicted while glucose is still below 180, so every alert is genuine early warning
- Refreshed every 15 minutes



Patient

Gets a timely nudge: “a 15-minute walk after lunch will likely keep you in range.”



Doctor

Sees a triage list of who is heading out of range, why, and what would help.



Care programme

Tracks time-in-range and targets coaching where it moves outcomes.

Glucotwin: fuse, simulate, predict

A hybrid of a personalised physiology model and machine learning



1 Fuse

Static EHR (labs, meds, genetics) and live wearables (CGM, HR, HRV, sleep, steps, meals) are aligned on one 5-minute timeline.



2 Simulate

A per-patient twin learns six physiological parameters in one week, then simulates the next 2 hours and answers what-if questions.



3 Predict & explain

Gradient boosting combines raw signals with the twin's simulation, outputs a calibrated risk and peak range, and shows the top reasons.

Two data streams, inside the sandbox rules

Fully synthetic cohort; no real patient data touched (DPDP Act / HIPAA)



Static / historical EHR

- 120 synthetic Indian T2D patients
- Demographics, diagnoses, HbA1c, fasting glucose, lipids, eGFR, BP
- Medications and TCF7L2 genotype
- FHIR-style export, LOINC / SNOMED codes; Synthea adapter included



Dynamic / real-time wearables

- CGM glucose every 5 minutes, 14 days
- Heart rate, HRV, steps, sleep stages
- Meal-log app; only ~70% of meals logged, with estimation error
- Physiology: sleep loss, walking, stress, dawn effect, rice vs wheat diets

483,840

sensor rows

56,744

prediction points

57

fused features

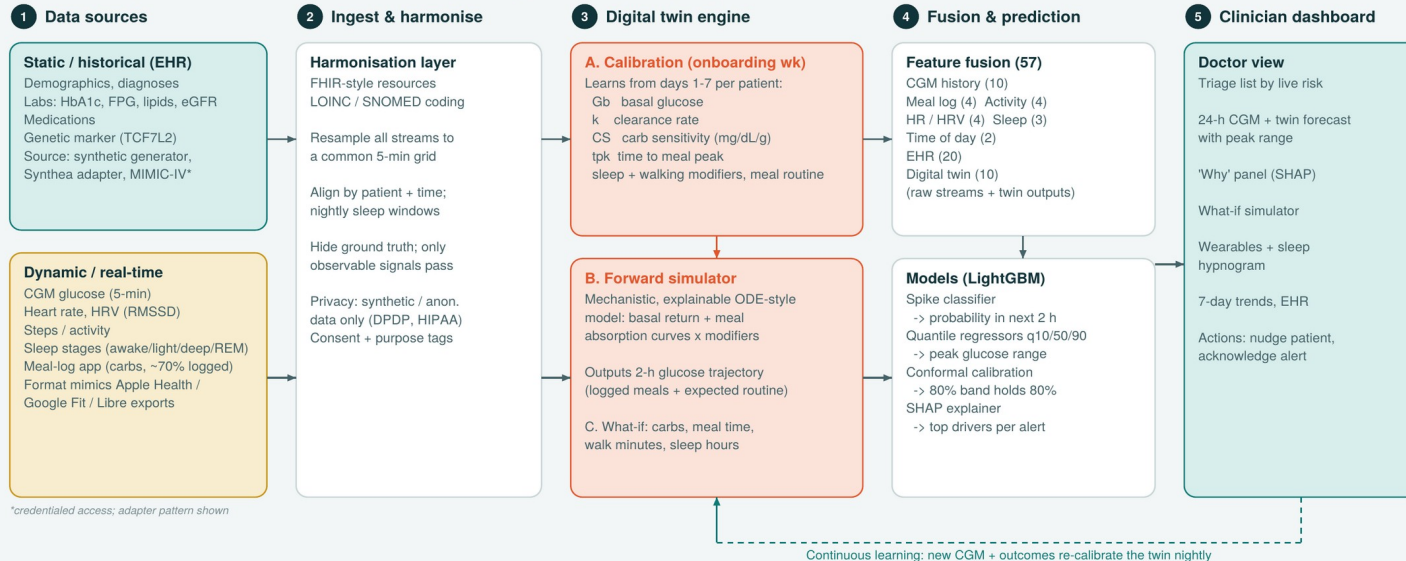
72%

cohort time-in-range

Architecture

Glucotwin: system architecture

Type 2 diabetes digital twin that fuses EHR + wearable streams to predict a glucose spike (>180 mg/dL) up to 2 hours ahead



Evaluation protocol

Patient-level split: test patients never seen in training (no leakage)
5-fold GroupKFold CV, ablation per stream, trend-arrow clinical baseline
Clinical metrics: detection rate, lead time, alert episodes/day, calibration

Production path (roadmap)

REST API (FastAPI) + message queue for device streams
ABDM / FHIR R4 integration for hospital EHRs; on-device CGM ingestion
Prospective validation with a clinical partner before any patient use

The twin learns each patient's physiology

Six interpretable parameters, calibrated from one onboarding week

Basal glucose

where glucose settles overnight

Carb sensitivity

mg/dL rise per gram of carbs

Time to peak

minutes from meal to peak

Clearance rate

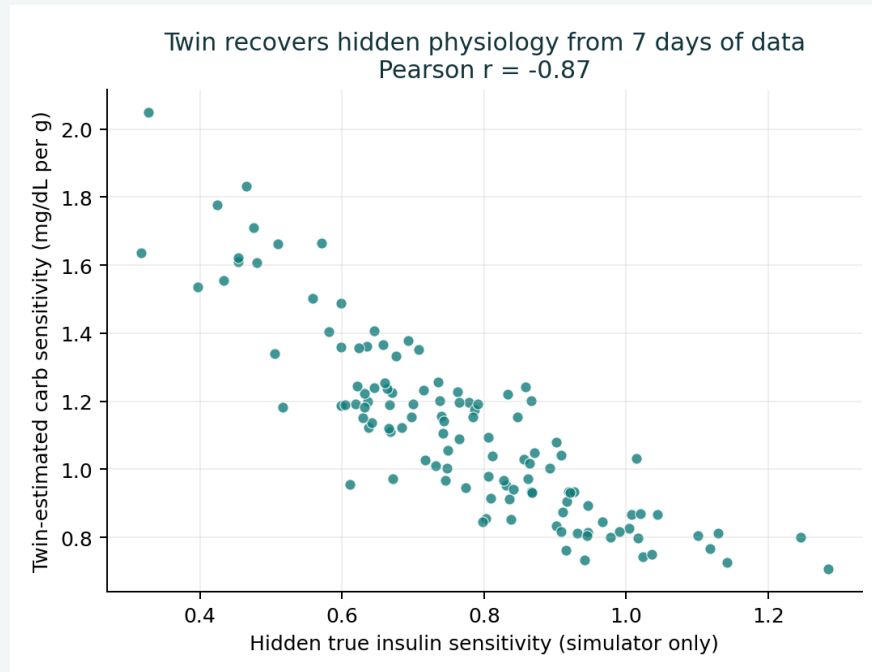
how fast glucose returns

Sleep sensitivity

extra rise per hour of lost sleep

Walking benefit

peak cut by a post-meal walk



Validated: the twin's carb sensitivity tracks the hidden true insulin sensitivity ($r = -0.87$).

Results on unseen patients

Patient-level split: 29 test patients never seen in training

0.934

AUROC

95%

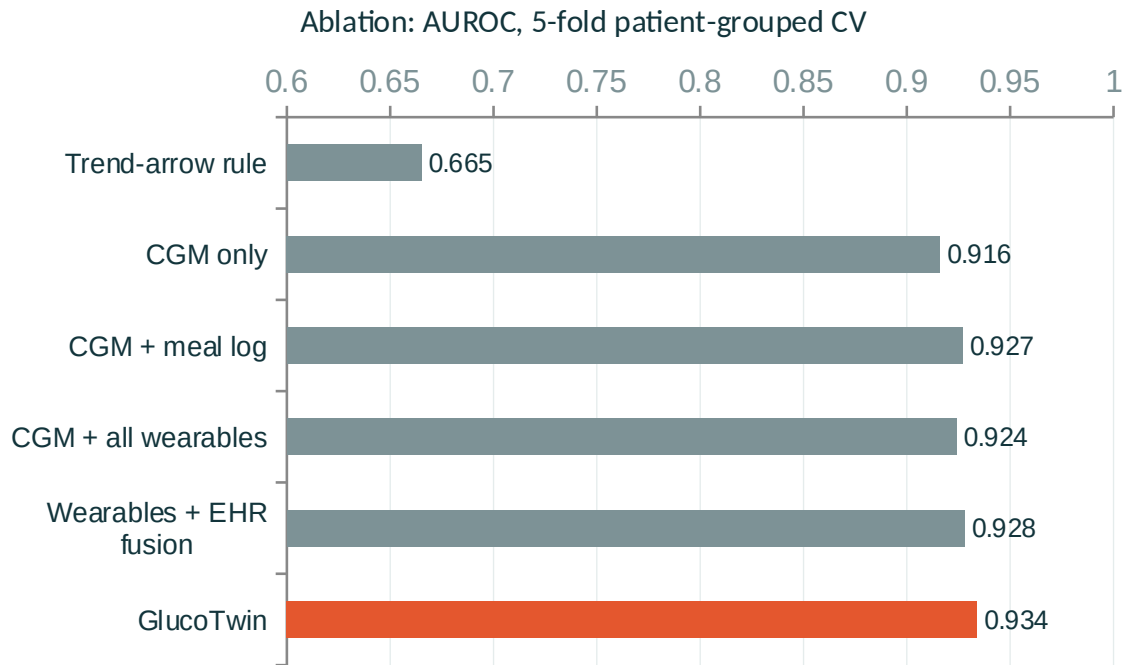
of spikes flagged in advance

90 min

median warning time

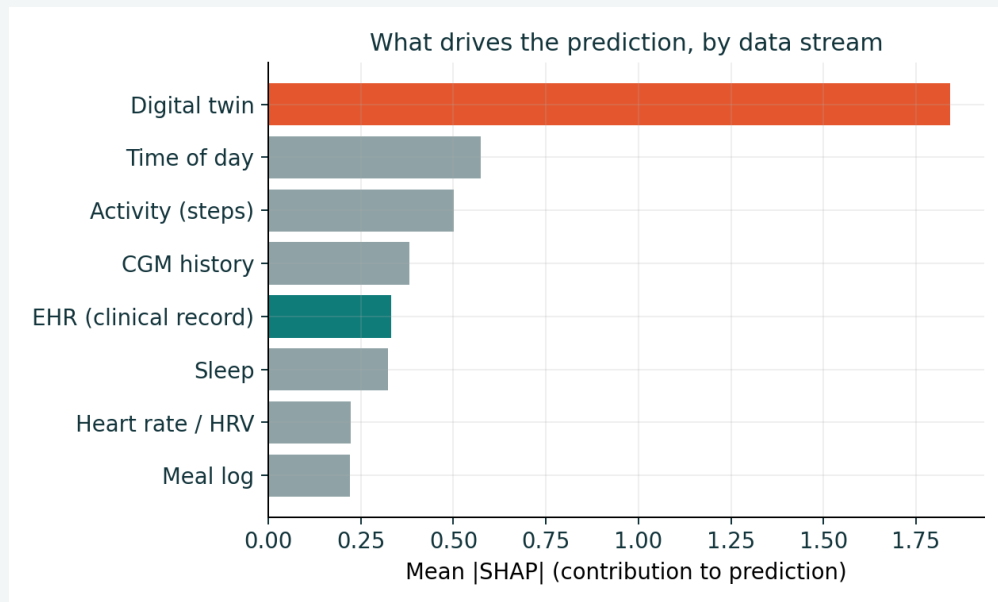
12 mg/dL

error on 2-h peak (vs 24 naive)



Every alert comes with its reasons

SHAP attributions, grouped by data stream and shown per prediction



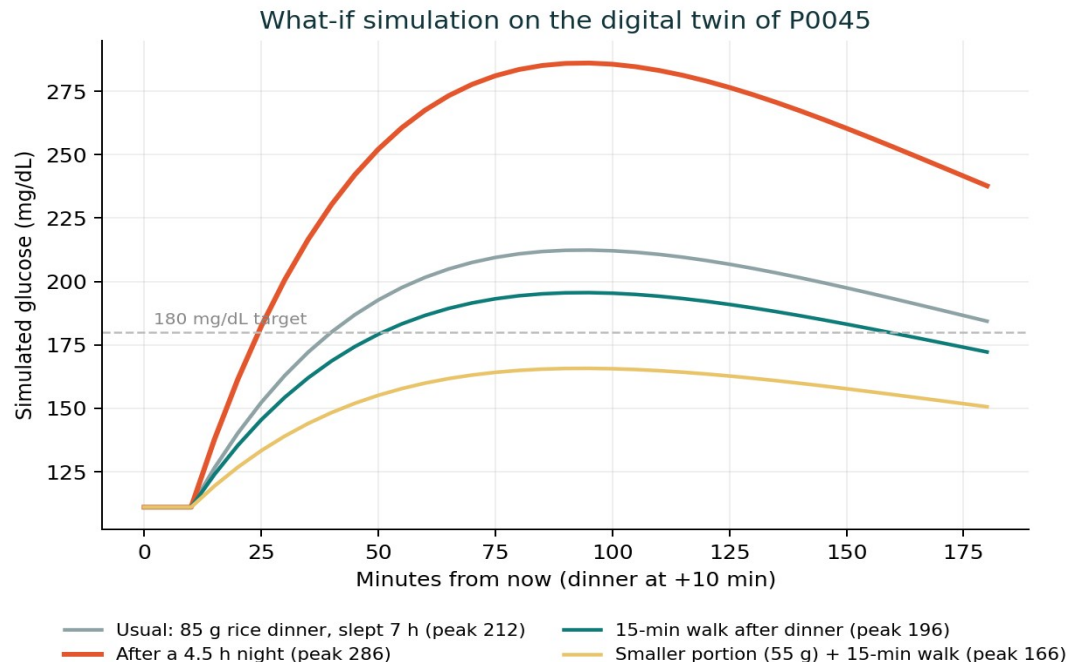
Fatima, 13:00: top reasons

- Twin-simulated peak: 252 mg/dL
- Usual lunch window (time of day)
- High personal basal glucose: 171
- 56% chance of a meal in next 2 h

Her lunch was never logged, yet she peaked at 259, inside the predicted 193–275 range. The twin knew her routine.

Ask the twin before advising the patient

Same 85 g rice dinner, simulated on patient P0045's twin



212 mg/dL

peak · Usual

286 mg/dL

peak · After a 4.5 h night

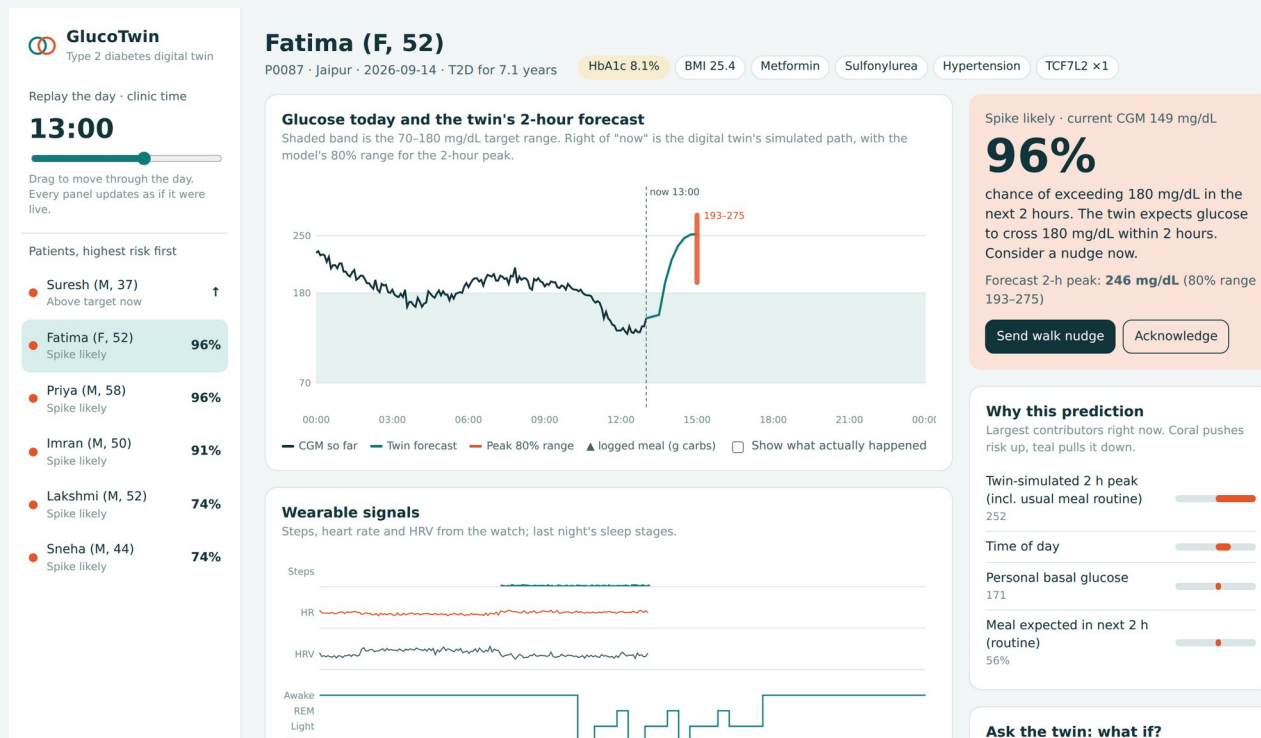
196 mg/dL

peak · 15-min walk after

166 mg/dL

peak · Smaller portion + walk

The clinician dashboard



Single HTML file, works offline

- Triage list re-sorts by live risk
- 24-h CGM with the twin's forecast and peak range
- Risk, reasons and one-tap patient nudge
- What-if sliders on the twin
- Replay the day to see alerts arrive in time

What this proves, what it doesn't, what's next



Limitations

- Synthetic data: metrics show the method works, not clinical accuracy
- The simulator and twin share a physiological family, which flatters results
- Relies on a CGM; meal logging is imperfect
- Not a medical device; decision support only



Roadmap

- Validate on open CGM + wearable datasets, then a prospective pilot with a diabetologist
- ABDM / FHIR R4 integration for hospital records
- Add hypoglycaemia risk for insulin and sulfonylurea users
- Sequence models once multi-month data exists
- Regional-language nudges via WhatsApp

Thank you

GlucoTwin sees the spike coming, explains why, and shows what would prevent it.

Stack: Python · LightGBM · SHAP · scikit-learn · pandas · HTML/SVG dashboard

Code, data generator, models, dashboard: [\[GitHub repository link\]](#)

Open source under the MIT License

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